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Drive Line Alignment for Single Bearing Alternators

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Engine models affected: K19, K38, K50, QST30, QSK19, QSK23, QSK38, QSK45, QSK50, QSK60, QSK78, and V28

Purpose

- The purpose of this service bulletin is to define the installation requirements and recommendations to optimize driveline alignment for single bearing electric drive systems, based on the misalignment capacity of Cummins® supplied flexplates and flywheels.

Installation Requirements

- Driveline Interface Requirements:
 - Axial misalignment between the engine/machine interface **must not** exceed 0.13 mm [0.005 in].
 - Radial misalignment between the engine/machine interface **must not** exceed 0.08 mm [0.003 in].
 - The process **must** be designed to minimize angular misalignment between the engine crankshaft and alternator shaft.
 - Maximum flexplate deflection **must not** exceed 1.27 mm [0.050 in] based in end clearance tolerances of the crankshaft and the alternator.

Component Interface Requirements:

- Installed flexplate face runout **must not** exceed:
 - 1 mm [0.040 inch] for SAE "00" flexplate flywheels, and
 - 0.89 mm [0.035 inch] for SAE "0" flexplate flywheels.

Installation Recommendations

- Driveline Interface Recommendation:
 - Axial end clearance of the generator **must** be less than the maximum crankshaft end clearance.

Introduction and Overview

- Alignment between the engine and driven component is a critical characteristic that **must** be evaluated during the machine assembly process. Proper driveline alignment can improve machine robustness and operator satisfaction, while improper alignment can lead to premature component malfunctions and operator discomfort.
- The three types of misalignment are axial, radial, and angular. See Figure 1 below.

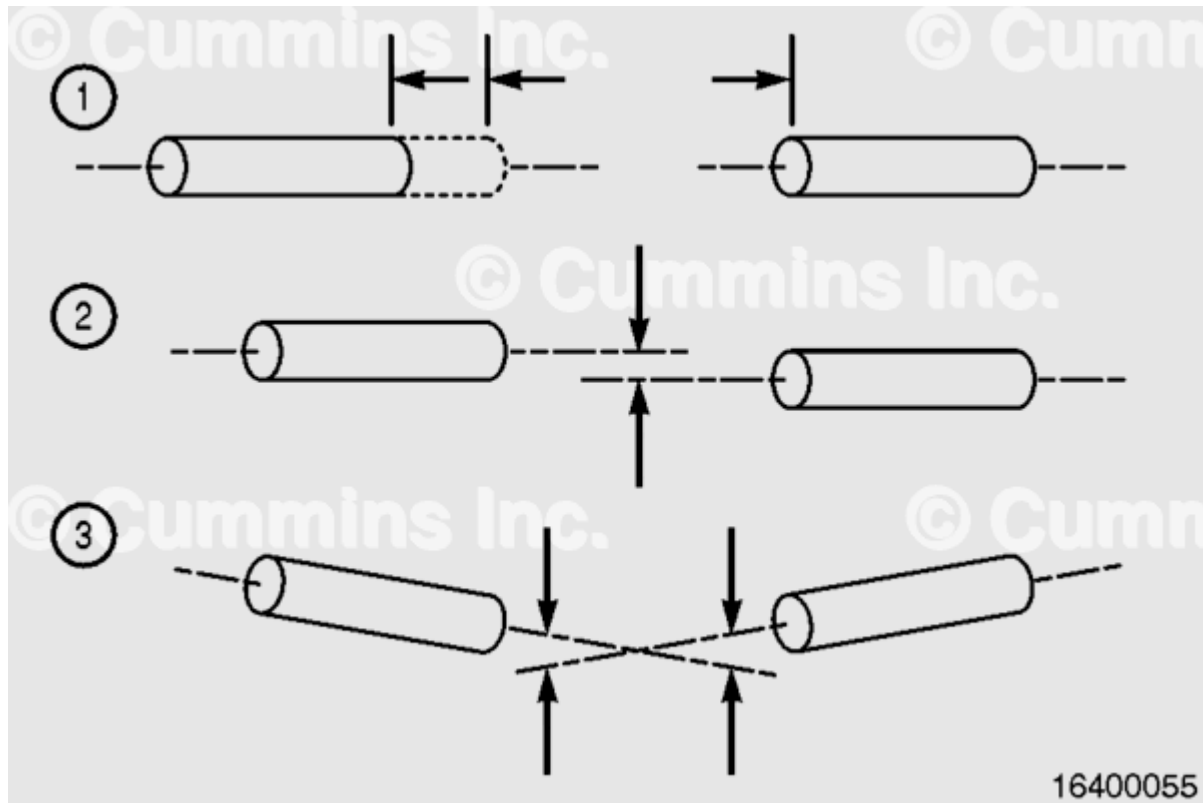


Figure 1: Types of Misalignment: Axial (1); Radial (2); Angular (3)

The effect of improper alignment propagates both ways from the flexible coupling interface. It can produce excessive strain on the engine in one direction and on the driven component (i.e. alternator) in the other direction. Symptoms of excessive misalignment may include:

- Damper malfunctions
- Flexplate deformation or cracking
- Flywheel cracking
- Bearing wear
- Bent or broken alternator drive shafts and crankshaft
- Engine mount malfunctions
- Excessive machine vibration leading to operator discomfort.

The following is a list of component dimensions and characteristics that **must** be considered and accounted for during the alignment process of an alternator.

Engine	Alternator
Crankshaft end clearance	Rotor shaft end clearance
Flywheel housing radial runout	Stator housing radial runout
Flywheel housing face runout	Stator housing face runout

Engine	Alternator
Flexplate face runout	Rotor adaptor face runout
Flexplate radial runout	Rotor adaptor radial runout

An Illustrated Method of Checking Engine Dimensions and Aligning Engine with Driven Attachments

- The following steps are used to verify proper engine assembly and component tolerances, as well as provide a "centered" crankshaft dimension used in axial alignment. The alternator manufacturer's service manual should be referred to for correct alternator end clearance and runout tolerances.

⚠ CAUTION ⚠

Do not strike or pry on the vibration dampers on the front of the engine.

Crankshaft End Clearance Measurement

- Move the crankshaft rearwards as far as possible. This can be accomplished by removing a hand-hole cover and prying gently on a crankshaft counter weight **or** using a rubber mallet on the front end of the crank.
- Measure and record the distance relative to the flywheel housing face to the mounting face of the flywheel/flexplate at four locations spaced 90 degrees apart, as shown in Figure 6. Record the average of these four measurements as Crank Rear Position (CRP).
- Move the crankshaft forward as far as possible by the same method.
- Measure and record the distance relative to the flywheel housing face to the mounting face of the flywheel/flexplate at four locations spaced 90 degrees apart, as shown in Figure 6. Record the average of these four measurements as Crank Forward Position (CFP).

Crankshaft end clearance = CFP - CRP.

The allowable range of endplay is shown in the table below:

Engine Model	Minimum End Clearance	Maximum End Clearance
K19, QSK19	0.10 mm [0.004 inch]	0.41 mm [0.016 inch]
QSK23	0.14 mm [0.006 inch]	0.32 mm [0.013 inch]
QST30	0.14 mm [0.006 inch]	0.35 mm [0.014 inch]
K38/K50	0.13 mm [0.005 inch]	0.51 mm [0.020 inch]
QSK45/QSK60	0.13 mm [0.005 inch]	0.51 mm [0.020 inch]
V28	0.15 mm [0.006 inch]	0.33 mm [0.013 inch]
QSK78	0.13 mm [0.005 inch]	0.52 mm [0.020 inch]

Flywheel Housing Face Runout

- Locate the dial indicator base of the machined face of the flywheel/flexplate and zero the probe on the flywheel housing face.
- Use the engine barring device to slowly rotate the engine crank through two full revolutions. Take note of the dial gauge reading during rotation. When complete, make sure the dial indicator returns to zero. The difference between the highest and lowest values is the face runout.

Limits are shown in the table below:

Flywheel Housing Face Runout	Maximum Value
Cummins® SAE "0" flywheel housing	0.25 mm [0.010 inch]
Cummins® SAE "00" flywheel housing	0.30 mm [0.012 inch]

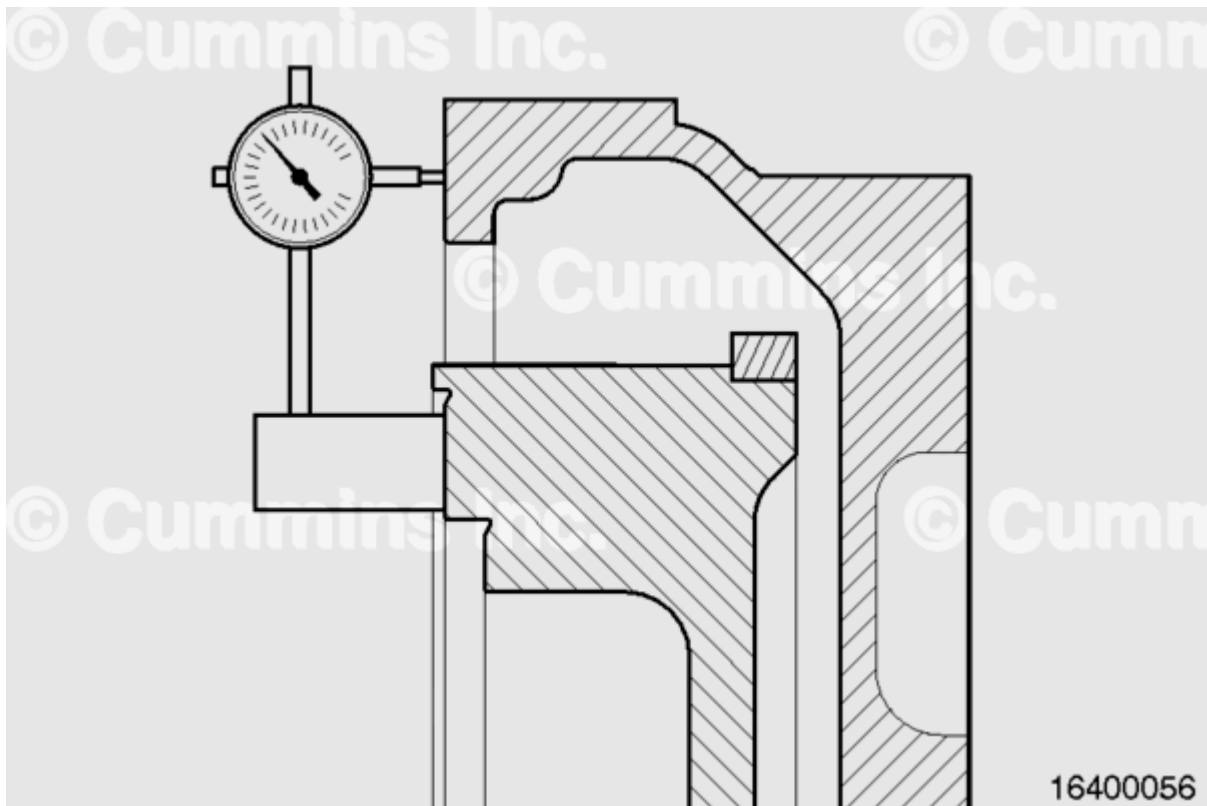


Figure 2: Flywheel Housing Face Runout Measurement

Flywheel Housing Radial Runout

- Locate the dial indicator base on the machined face of the flywheel/flexplate and zero the probe on the inside diameter of the pilot bore of the flywheel housing.
- Use the engine barring device to slowly rotate the engine crankshaft through two full revolutions. When complete, make sure the dial indicator returns to zero. The difference

between the highest and lowest values is the flywheel housing radial runout.

Limits are shown in the table below:

Flywheel Housing Radial Runout	Maximum Value
Cummins® SAE "0" flywheel housing	0.25 mm [0.010 inch]
Cummins® SAE "00" flywheel housing	0.30 mm [0.012 inch]

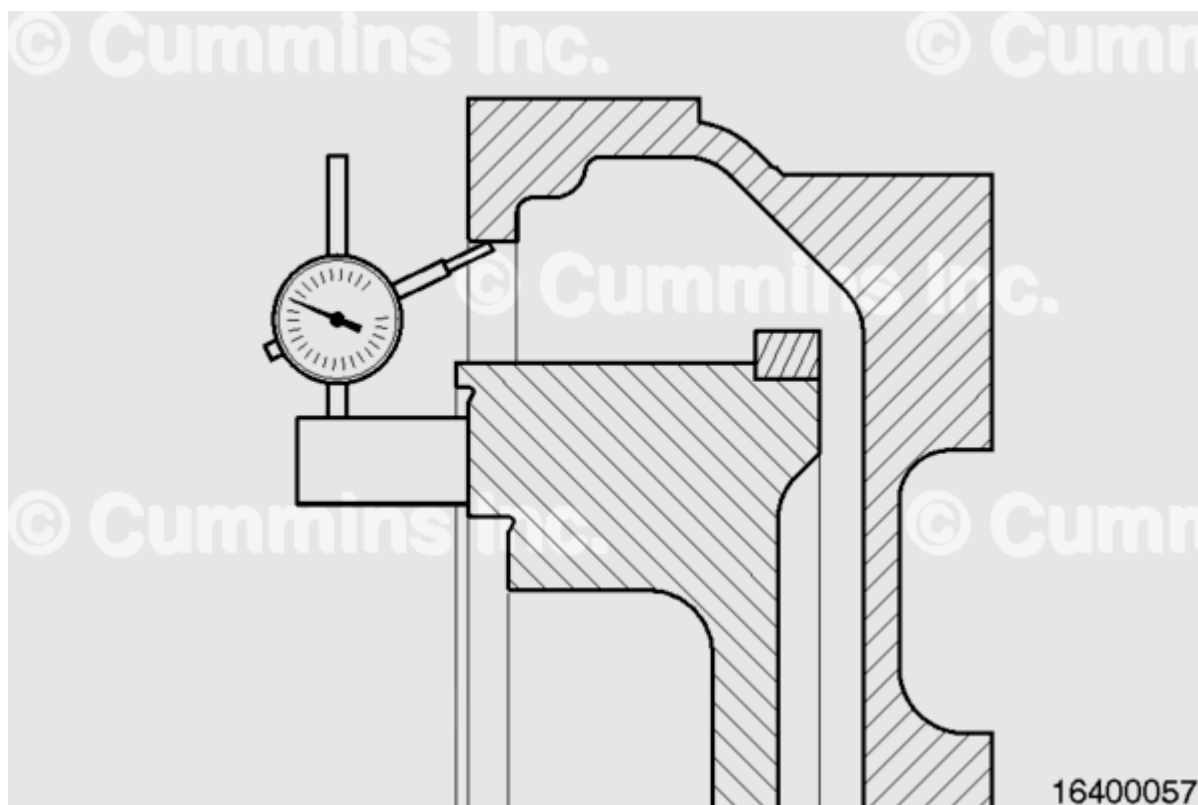


Figure 3: Flywheel Housing Radial Runout Measurement

Flexplate Face Runout

- Locate the dial indicator base on the machined face of the flywheel housing and zero the probe on the face of the flywheel/flexplate.
- Use the engine barring device to slowly rotate the engine crankshaft through two full revolutions. When complete, make sure the dial indicator returns to zero. The difference between the highest and lowest values is the flexplate face runout.

Limits are shown in the table below:

Flexplate Face Runout	Maximum Value
Cummins® SAE "0" flexplate	0.89 mm [0.035 inch]
Cummins® SAE "00" flexplate	1.00 mm [0.040 inch]

Note : If the maximum value is exceeded, the flexplate must be replaced.

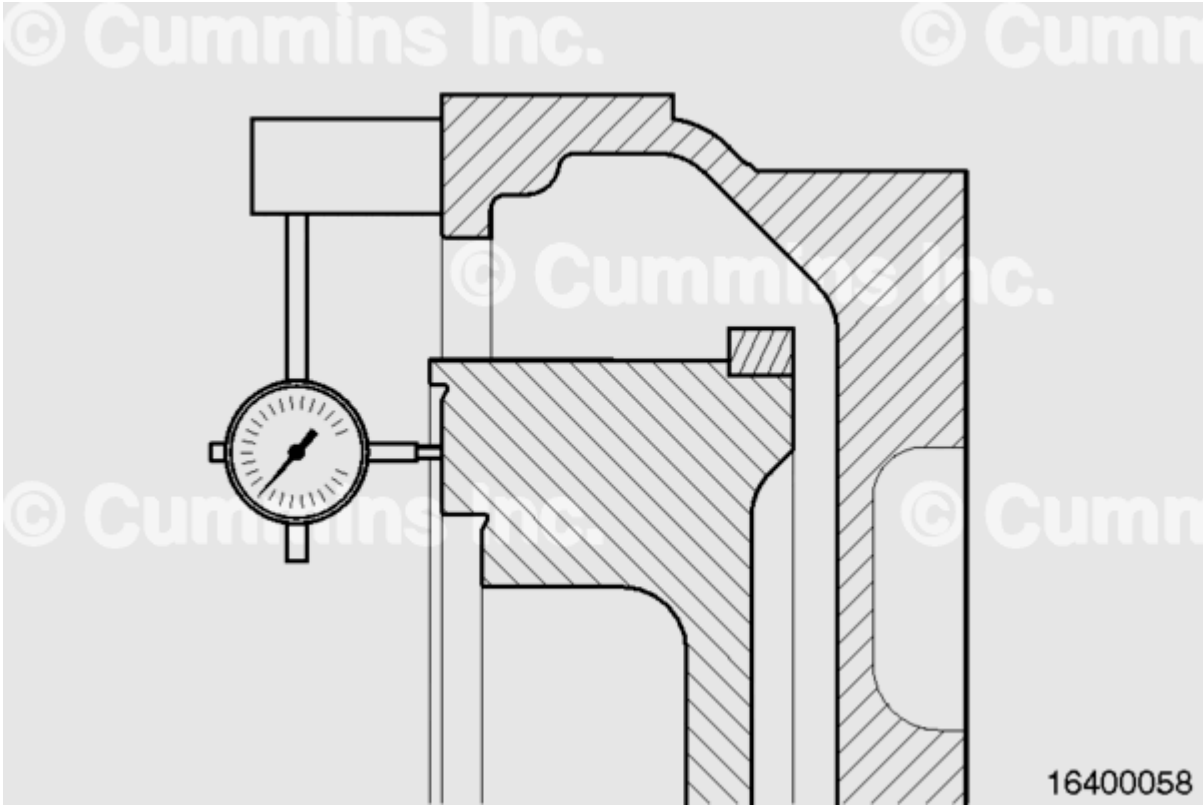


Figure 4: Flywheel/Flexplate Face Runout Measurement

Flexplate Radial Runout

- Locate the dial indicator base on the machined face of the flywheel housing and zero the probe on the inside diameter of the flywheel/flexplate pilot bore.
- Use the engine barring device to slowly rotate the engine crankshaft through 2 full revolutions. When complete, make sure the dial indicator returns to zero. The difference between the highest and lowest values is the flexplate radial runout.

Limits are shown in the table below:

Flexplate Radial Runout	Maximum Value
Cummins® SAE "0" flywheel/flexplate	0.08 mm [0.003 inch]
Cummins® SAE "00" flywheel/flexplate	0.13 mm [0.005 inch]

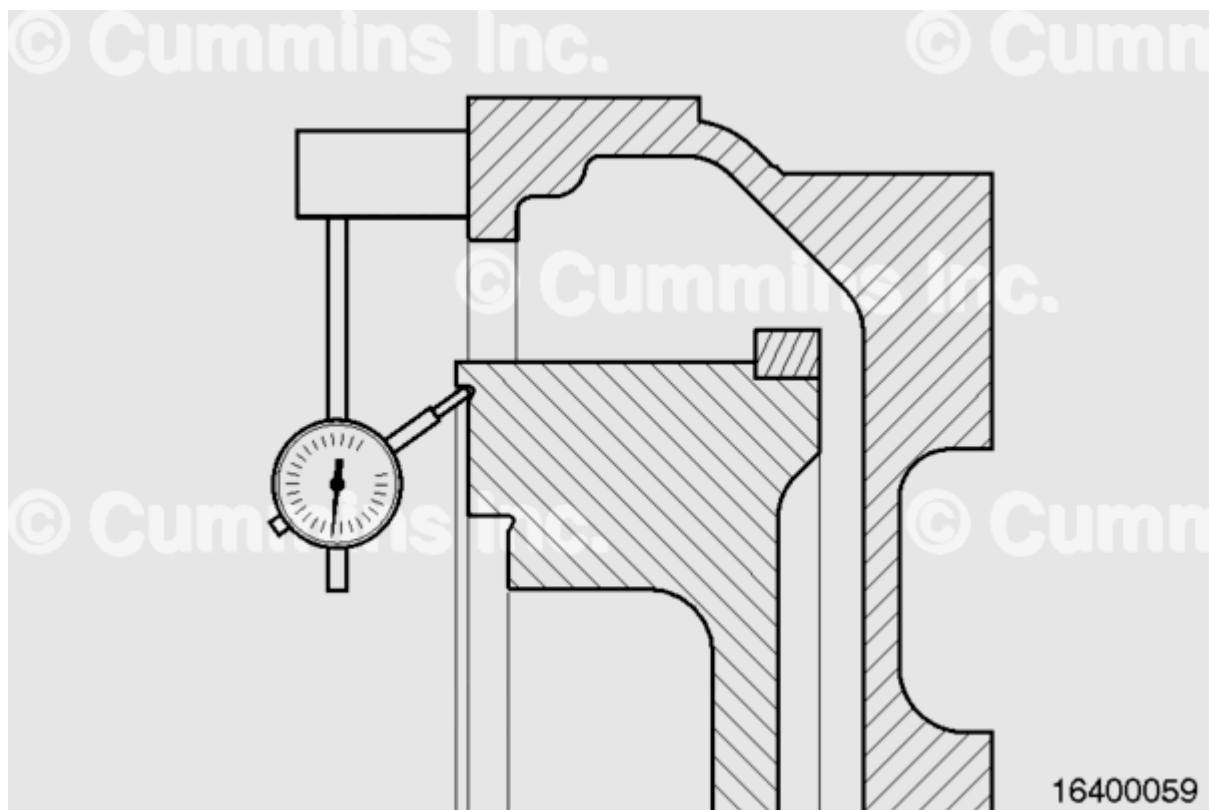


Figure 5: Flywheel/Flexplate Radial Runout Measurement

Nominal Flywheel/Flexplate Position

- The centered, or nominal, crankshaft position (CNP) is the dimension from the flywheel housing face to the flywheel/flexplate face, when the crank is positioned in the center of its endplay. CNP can be calculated using the formula:
 - $CNP = [CRP + CFP] \times 0.5$
- Use the values obtained from the crankshaft end clearance measurement.
- If CNP is positive, the flywheel/flexplate mounting face is inside the flywheel housing mounting face. If it is negative, the flywheel/flexplate mounting face is outside the flywheel housing mounting face.

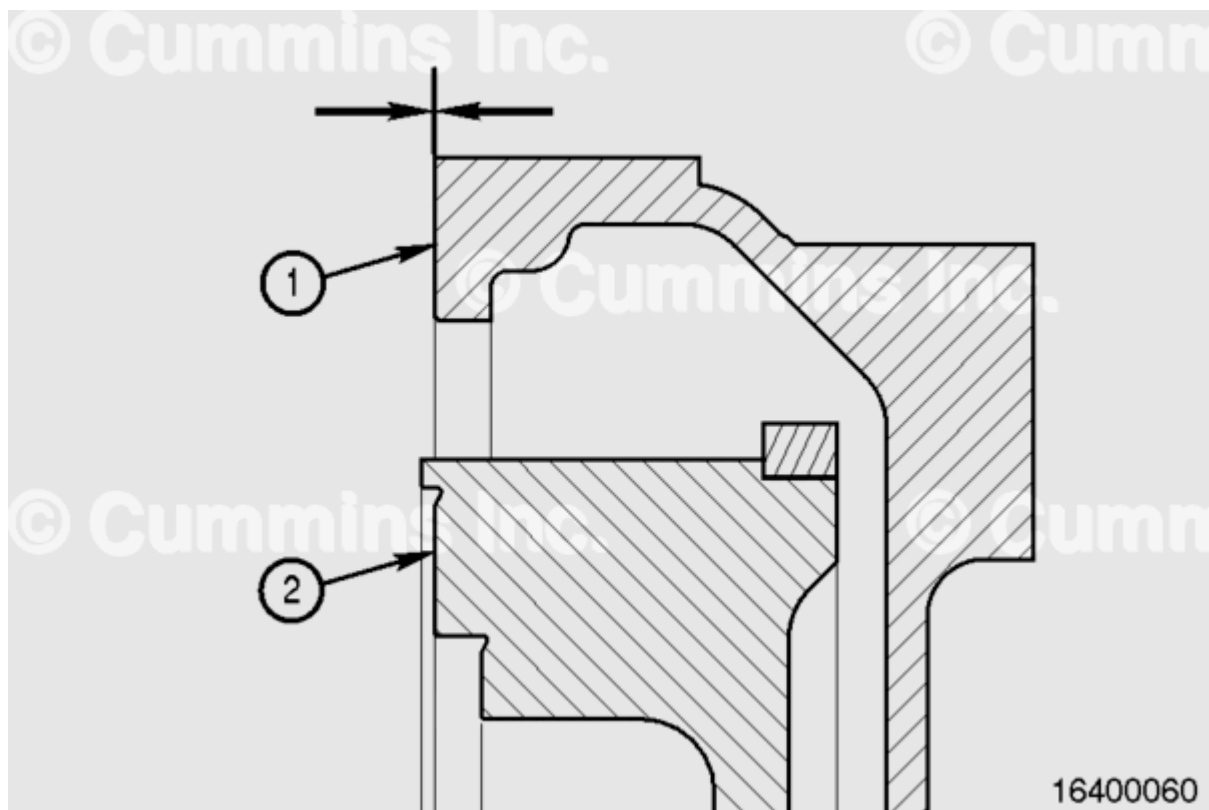


Figure 6: Nominal Flywheel/Flexplate Position

Attachment Nominal Position

- Figure 7 illustrates a representative attachment to the QSK60 SAE 00 flywheel. It illustrates that the ANP is the measured distance between the alternator housing face and the driveplate face.

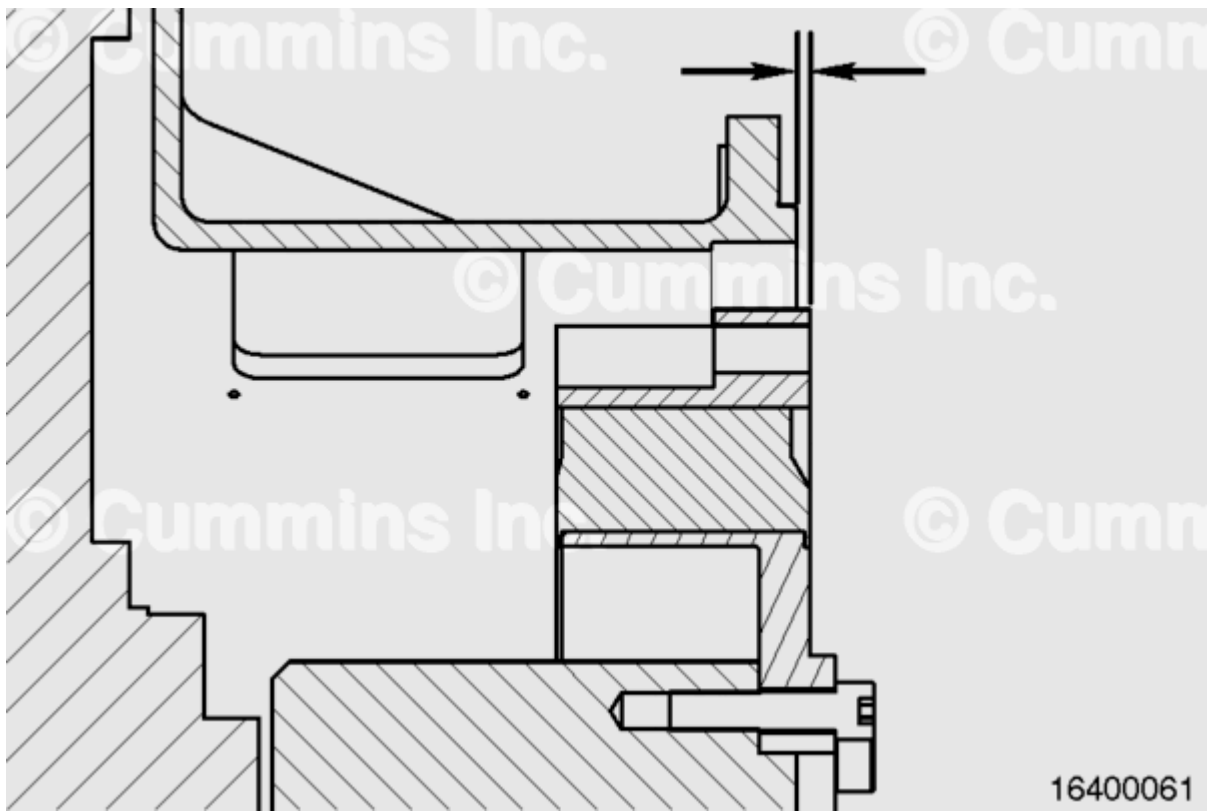


Figure 7: Attachment Nominal Position

It is critical that the attachment nominal position (ANP) be within ± 0.13 mm [0.005 in] to prevent overstressing the flexible element of the driveline, either a Cummins® flexplate or the attachment flexible element.

Attachment Nominal Position	Maximum Value
Cummins® SAE "0" flywheel/flexplate	-CNP ± 0.13 mm [0.005 inch]
Cummins® SAE "00" flywheel/flexplate	-CNP ± 0.13 mm [0.005 inch]

Attachment Shaft Runout and Radial Alignment

- See Figure 8 below for a representative illustration of the concepts discussed here. Install and tighten the bolts attaching the driven member to the engine flywheel housing. Attach at least four bolts from the driven member to the flywheel/flexplate and tighten.

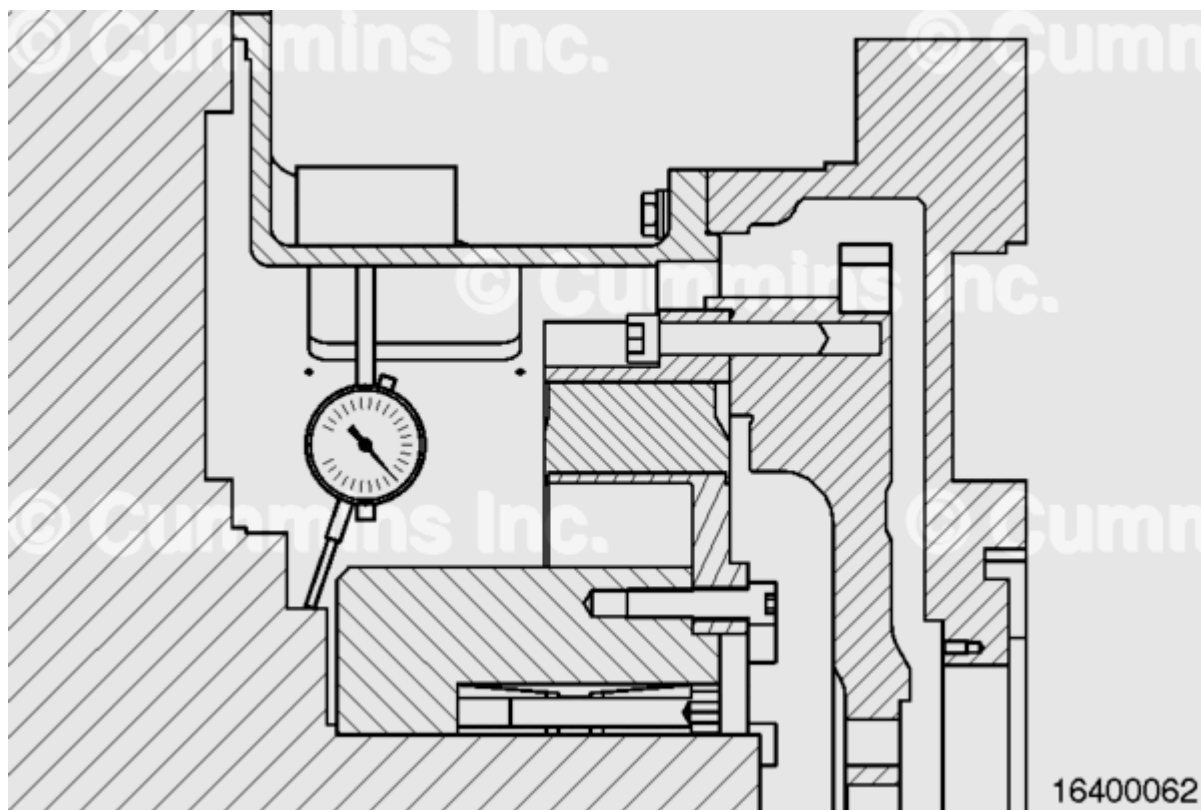


Figure 8: Attachment Shaft Runout and Radial Alignment

- Locate the base of the dial indicator on the stationary portion of the attachment. The probe should be zeroed on a clean, machined surface of the shaft to measure radial runout.
- Use the engine barring device to slowly rotate the engine crankshaft through 2 full revolutions. When complete, make sure the dial indicator returns to zero. The difference between the maximum and minimum values while rotating is the assembled radial runout.
- Assembled radial runout maximum value: 0.08 mm [0.003 in]
- If measured assembled radial runout is less than the maximum allowed value, install and torque the remaining bolts attaching the driven member to the flywheel/flexplate and flywheel housing. Check the assembled radial runout to make sure it is still less than the maximum allowed value.
- If assembled radial runout is greater than the maximum value, perform the following steps:
 1. Determine the point on the shaft with the greatest radial runout.
 2. Place this at the top, or twelve o' clock position.
 3. Loosen the bolts attaching the driven member to the flywheel/flexplate completely.
 4. Tighten the flywheel/flexplate bolts. This should allow the driven member to "center" itself and reduce runout.
 5. Measure the radial runout. If the measurement is less than the maximum allowed value, install and torque the remainder of the flywheel/flexplate attachment bolts and check the radial runout.
 6. If the measurement is greater than the maximum allowed value, repeat steps 1 through 5, OR remove the flywheel bolts and, while holding the attachment stationary, rotate the engine crankshaft 90 degrees and attach the driven member to the flywheel/flexplate and repeat steps 1 through 5.

Document History

Date	Details
14-03-2017	Module Converted
2017-4-4	Update to Format
2017-5-4	Corrected table headings in Flexplate Face and Radial Runout tables. Also converted legacy tagging.

Last Modified: 04-May-2017
